

# Free Energy Is a Signed, Zero-Training Detector of Acquisition Shift in Frozen Vision–Language Encoders: A Pre-Registered Falsification Study on Medical Image Corruptions

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## Abstract

Frozen vision–language encoders such as CLIP enable zero-shot medical image classification in which class anchors are text embeddings and the softmax posterior over image–anchor distances is exactly a Gibbs distribution at temperature  $T = 2\tau$ , where  $\tau$  is the temperature learned during contrastive pre-training. This identification makes two label-free observables available at no training cost: the Shannon entropy  $\mathcal{H}$  of the posterior and the Helmholtz free energy  $F$  of the underlying energy model. Because a uniform translation of the anchor distances leaves  $\mathcal{H}$  invariant but shifts  $F$ , we pre-registered a falsifiable prediction: under increasing acquisition shift,  $F$  separates source from corrupted images ( $\text{AUROC} \geq 0.70$ ) while  $\mathcal{H}$  stays near chance ( $\text{AUROC} \leq 0.60$ ). We tested this prediction with a strictly zero-training protocol—prompts frozen before any target evaluation, temperature fixed at  $2\tau$ , every reported number traceable to a versioned run—on the official MedMNIST-C corruptions of BreastMNIST (ultrasound, seven corruptions) and PneumoniaMNIST (chest X-ray, four corruptions), with paired DeLong tests and stratified bootstrap confidence intervals. The strong form of the prediction is *refuted*: entropy is never near chance. What the data reveal instead is sharper. On structural corruptions (pixelation, JPEG, motion blur, speckle)  $F$  dominates  $\mathcal{H}$  significantly (severity-5 AUROC up to 1.000 vs. 0.888; paired  $\Delta\text{AUROC} +0.080$  to  $+0.208$ , all  $p < 0.05$ ), whereas photometric corruptions (brightness, contrast, X-ray Gaussian noise) *decrease*  $F$  and drive the one-sided detector below or near chance (AUROC 0.127–0.594): free energy is a *signed* shift score. A two-sided variant  $|F - \text{med}_{\text{source}}(F)|$ , calibrated on source images only, restores above-chance detection on five of six sign-inverted cases (e.g.,  $0.127 \rightarrow 0.803$ ) while preserving the structural ones. Across prompt reformulations  $F$  is markedly more stable than  $\mathcal{H}$  (severity-5 AUROC dispersion 0.03 vs. 0.08). All code, per-run metrics and raw scores are released; refuted criteria are reported as pre-registered.

## 1 Introduction

Zero-shot classification with frozen vision–language encoders (Radford et al., 2021) is an attractive substrate for medical imaging: class definitions are text prompts, no gradient touches the backbone, and the deployment surface is a single forward pass. Yet clinical images arrive from scanners, probes and acquisition protocols that drift over time and across sites, and a deployed zero-shot pipeline needs a label-free signal that the input distribution has moved (Rabanser et al., 2019). Such a signal is the gating quantity for every downstream mechanism studied in test-time adaptation (TTA) (Wang et al., 2021; Shu et al., 2022; Yoon et al., 2024; Niu et al., 2022, 2023): whether to adapt, when to stop, and which samples to refer to a human reader.

This paper studies the cheapest conceivable such signal. When image embeddings  $\mathbf{z}$  are scored against  $K$  fixed text anchors  $\boldsymbol{\mu}_k$ , the CLIP posterior is a Gibbs (Boltzmann) distribution over squared distances at temperature  $T = 2\tau$  (Section 2). Two observables then come for free: the prediction entropy  $\mathcal{H}$ —the workhorse of TTA objectives and confidence baselines (Hendrycks and Gimpel, 2017; Wang et al., 2021)—and the Helmholtz free energy  $F$ , the log-partition function popularized for out-of-distribution detection on classifier logits (Liu et al., 2020; Grathwohl et al., 2020; LeCun et al., 2006) and later transposed to vision–language models (Ming et al., 2022).

Our starting point is a structural asymmetry between the two. If an acquisition shift moves images away from *all* anchors roughly uniformly — adding a constant  $c$  to every energy  $E_k$  — the posterior and hence  $\mathcal{H}$  are unchanged, but  $F$  shifts by exactly  $c$ . This motivated a *pre-registered, falsifiable* prediction (criteria fixed before any target data were scored, Section 4): under increasing corruption severity,  $F$  detects the shift (AUROC  $\geq 0.70$ ) while  $\mathcal{H}$  remains near chance (AUROC  $\leq 0.60$ ), with a margin of at least 0.05.

We tested this prediction under a deliberately austere protocol: no training, no adaptation, prompts frozen *a priori* and published verbatim, temperature fixed at the theoretical value  $2\tau$ , and every number in this paper generated by a versioned script and stored in a per-run metrics file. The testbed is MedMNIST-C (Di Salvo et al., 2024), the corruption benchmark built on MedMNIST v2 (Yang et al., 2023), on two modalities: breast ultrasound (BreastMNIST, derived from BUSI (Al-Dhabyani et al., 2020)) and chest X-ray (PneumoniaMNIST, derived from Kermany et al., 2018).

**Findings.** The strong form of the prediction is *refuted*:  $\mathcal{H}$  is never near chance (severity-5 AUROC 0.462–0.888), because realistic corruptions do not translate energies uniformly. The refutation, however, uncovers a structure that we believe is more useful than the original conjecture:

1. **A structural/photometric dichotomy.** On corruptions that destroy spatial structure (pixelation, JPEG compression, motion blur, speckle noise),  $F$  dominates  $\mathcal{H}$  significantly as a shift detector. On photometric corruptions (brightness, contrast, X-ray Gaussian noise), corrupted embeddings move *closer* to the text anchors,  $F$  decreases, and the one-sided detector drops below chance — to 0.127 in the worst case. Free energy is a *signed* score, not a magnitude.
2. **A source-calibrated two-sided rescue.** The statistic  $|F - \text{med}_{\text{src}}(F)|$ , using only the source median, restores above-chance detection on five of the six sign-inverted cases while leaving structural detection essentially intact.
3. **Prompt stability.** Across three prompt reformulations, the severity-5 detection AUROC of  $F$  varies by  $\sim 0.03$  while that of  $\mathcal{H}$  varies by  $\sim 0.08$ : the partition function is more robust to the wording that defines the anchors than the posterior shape is.

We report the pre-registered verdict in full, including the criteria that failed, and release code, metrics and raw scores for every run.

## 2 Theoretical framework

### 2.1 Gibbs posterior over textual anchors

Let  $\mathbf{z} \in \mathbb{R}^d$  be an  $\ell_2$ -normalized image embedding and  $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K$  the  $\ell_2$ -normalized text embeddings of the class prompts. Define the configuration energy

$$E_k(\mathbf{z}) = \|\mathbf{z} - \boldsymbol{\mu}_k\|_2^2 = 2 - 2\langle \mathbf{z}, \boldsymbol{\mu}_k \rangle. \quad (1)$$

The Gibbs distribution at temperature  $T > 0$  is  $p_k(\mathbf{z}) = \exp(-E_k/T) / \sum_j \exp(-E_j/T)$ . CLIP computes logits  $\langle \mathbf{z}, \boldsymbol{\mu}_k \rangle / \tau$  with a learned temperature  $\tau$ ; substituting (1) shows the two softmaxes coincide exactly when

$$T = 2\tau. \quad (2)$$

No temperature is tuned in this work:  $T$  is read off the frozen model (here  $T \approx 0.0200$ ).

## 2.2 Observables and the thermodynamic identity

We track the entropy, free energy, mean energy and a normalized order parameter (“crystallinity”):

$$\mathcal{H}(\mathbf{z}) = -\sum_k p_k \log p_k, \quad F(\mathbf{z}) = -T \log \sum_k e^{-E_k/T}, \quad \langle E \rangle(\mathbf{z}) = \sum_k p_k E_k, \quad (3)$$

$$\chi(\mathbf{z}) = 1 - \mathcal{H}(\mathbf{z}) / \log K, \quad \text{with the identity} \quad F = \langle E \rangle - T \mathcal{H} \quad (4)$$

used as a software invariant (unit-tested to numerical precision).  $F$  is, up to the distance parameterization, the negative energy score of Liu et al. (2020) evaluated on distance logits.

## 2.3 The pre-registered prediction

**Proposition 1** (Translation asymmetry). *For any constant  $c$ , the map  $E_k \mapsto E_k + c$  leaves  $p$  and therefore  $\mathcal{H}$  and  $\chi$  unchanged, while  $F \mapsto F + c$ .*

If acquisition shift pushed embeddings away from all anchors while roughly preserving relative class separations,  $F$  would detect it and  $\mathcal{H}$  would not. Before scoring any corrupted image we therefore froze the following decision criteria, evaluated at maximum severity with the source cohort as negatives:

- C1.  $\text{AUROC}(F) \geq 0.70$ ;
- C2.  $\text{AUROC}(\mathcal{H}) \leq 0.60$ ;
- C3.  $\text{AUROC}(F) - \text{AUROC}(\mathcal{H}) \geq 0.05$ .

The prediction is *confirmed* only if all three hold. These thresholds are decision rules, not results; a cleanly executed refutation is reported as such.

## 3 Related work

**Energy-based OOD detection.** Liu et al. (2020) showed that the logit log-partition function outperforms maximum softmax probability (Hendrycks and Gimpel, 2017) for OOD detection; Grathwohl et al. (2020) read classifiers as energy-based models. Ming et al. (2022) transposed confidence-based OOD scores to CLIP. Our contribution is orthogonal to score engineering: a pre-registered, severity-resolved comparison of  $F$  against  $\mathcal{H}$  on *medical acquisition-style* corruptions, exposing the signed behavior of  $F$  that global OOD benchmarks average away.

**Dataset-shift detection.** Rabanser et al. (2019) compared two-sample tests on learned representations. We restrict attention to the two statistics that are free in a deployed zero-shot pipeline, and quantify their per-sample discriminative power (AUROC) rather than cohort-level test power.

**Test-time adaptation.** Entropy minimization is the dominant TTA objective (Wang et al., 2021; Niu et al., 2022, 2023), extended to prompt tuning in vision-language models (Shu et al., 2022) with calibration corrections (Yoon et al., 2024). Our results bear on the *gating* of such methods: a signal that moves under shift while entropy barely moves (or moves for the wrong reason) is the natural trigger and stopping observable. This paper establishes the zero-training baseline for that program; the adaptation stage itself is out of scope here.

**Benchmarks.** MedMNIST-C (Di Salvo et al., 2024) provides modality-specific corruptions at five severities for MedMNIST v2 (Yang et al., 2023); we use its official corruption operators verbatim.

## 4 Experimental protocol

### 4.1 Substrate

Encoder: CLIP ViT-B/16 (OpenAI weights, QuickGELU variant) via `open_clip` (Ilharco et al., 2021), entirely frozen, run in float32 on CPU (PyTorch 2.4.1, Python 3.11, Windows). Temperature fixed at  $T = 2\tau \approx 0.0200$  per (2). Anchors are the  $\ell_2$ -normalized means of the per-class prompt-template embeddings (prompt ensembling), renormalized after averaging.

### 4.2 Cohorts and corruptions

**BreastMNIST** (Yang et al., 2023; Al-Dhabyani et al., 2020): breast ultrasound, binary malignant vs. normal/benign, test split  $n = 156$  at  $224 \times 224$  (MedMNIST+). All seven official MedMNIST-C corruptions for this dataset: `speckle_noise`, `motion_blur`, `brightness_up`, `brightness_down`, `contrast_down`, `jpeg_compression`, `pixelate`, at severities 0 (clean source) through 5. **PneumoniaMNIST** (Kermany et al., 2018): chest X-ray, binary normal vs. pneumonia, test split  $n = 624$ ; corruptions `gaussian_noise`, `gaussian_blur`, `brightness_down`, `contrast_down`.

Zero-shot diagnostic performance on clean data is deliberately reported and modest — BreastMNIST balanced accuracy 0.517 (near chance; accuracy 0.712, AUROC 0.636), PneumoniaMNIST accuracy 0.620 — the shift detection question is independent of diagnostic quality, and no claim of clinical utility of the zero-shot classifier is made.

### 4.3 Prompts, frozen a priori

The prompt sets were fixed before any corrupted image was scored and are reproduced verbatim. BreastMNIST classes {"a malignant breast tumor", "a normal or benign breast tissue"} with templates {"a breast ultrasound image of {}.", "an ultrasound scan showing {}.", "grayscale sonography of {}."}; PneumoniaMNIST classes {"a normal chest", "a chest with pneumonia"} with templates {"a chest x-ray image of {}.", "a radiograph showing {}."}. Prompt sensitivity is measured by re-running the full protocol with each single template (Section 5.4).

### 4.4 Metrics and statistics

For each severity  $s \geq 1$  we compute the AUROC of  $F$  (and of  $\mathcal{H}$ ) as a per-sample discriminator between the clean cohort ( $s = 0$ , negatives) and the corrupted cohort (positives); higher score should indicate the corrupted domain. Because both scores are computed on the same samples, we compare them with the paired DeLong test (DeLong et al., 1988) and a stratified paired bootstrap (Efron and Tibshirani, 1994) with  $10^4$  resamples for 95% confidence intervals on  $\Delta\text{AUROC} = \text{AUROC}(F) - \text{AUROC}(\mathcal{H})$ . The two-sided variant replaces a sample’s score  $x$  by

Table 1: Severity-5 shift detection (source vs. corrupted), canonical runs (prompt ensemble, seed 0, severities 0–5).  $\Delta = \text{AUROC}(F) - \text{AUROC}(\mathcal{H})$  with stratified paired bootstrap 95% CI and paired DeLong  $p$ -value.  $|F|_2 = \text{two-sided variant } |F - \text{med}_{\text{src}}(F)|$ . Structural corruptions above the rule, photometric below.

| Dataset   | Corruption       | AUROC( $F$ ) | AUROC( $\mathcal{H}$ ) | $\Delta$ [95% CI]       | $p$                    | $ F _2$ |
|-----------|------------------|--------------|------------------------|-------------------------|------------------------|---------|
| Breast    | pixelate         | 1.000        | 0.888                  | +0.112 [+0.078, +0.150] | $1.9 \times 10^{-9}$   | 1.000   |
| Breast    | jpeg_compression | 0.920        | 0.711                  | +0.208 [+0.143, +0.274] | $7.7 \times 10^{-10}$  | 0.899   |
| Breast    | motion_blur      | 0.885        | 0.760                  | +0.125 [+0.060, +0.190] | $1.7 \times 10^{-4}$   | 0.845   |
| Breast    | speckle_noise    | 0.814        | 0.734                  | +0.080 [+0.006, +0.154] | 0.032                  | 0.742   |
| Pneumonia | gaussian_blur    | 0.970        | 0.798                  | +0.172 [+0.148, +0.198] | $4.1 \times 10^{-41}$  | 0.962   |
| Breast    | brightness_up    | 0.594        | 0.677                  | −0.083 [−0.176, +0.012] | 0.083                  | 0.447   |
| Breast    | brightness_down  | 0.359        | 0.563                  | −0.204 [−0.297, −0.112] | $1.3 \times 10^{-5}$   | 0.606   |
| Breast    | contrast_down    | 0.299        | 0.498                  | −0.199 [−0.287, −0.112] | $7.4 \times 10^{-6}$   | 0.630   |
| Pneumonia | gaussian_noise   | 0.127        | 0.462                  | −0.335 [−0.368, −0.300] | $1.7 \times 10^{-86}$  | 0.803   |
| Pneumonia | brightness_down  | 0.200        | 0.644                  | −0.445 [−0.482, −0.406] | $8.0 \times 10^{-118}$ | 0.598   |
| Pneumonia | contrast_down    | 0.209        | 0.698                  | −0.490 [−0.526, −0.452] | $1.7 \times 10^{-148}$ | 0.589   |

$|x - \text{med}_{\text{src}}|$  where  $\text{med}_{\text{src}}$  is the median of the *source* scores; no target information enters the calibration.

#### 4.5 Leakage controls and traceability

(i) Prompts frozen before target evaluation; (ii)  $T = 2\tau$ , never tuned; (iii) no threshold is fit on target data (the two-sided center uses source only); (iv) every number below carries a run identifier; each run stores its configuration, git commit, code hash, environment, per-severity metrics and raw per-sample scores. A caveat we cannot close: CLIP’s web-scale pre-training corpus is not auditable, so memorization of the public source images cannot be excluded; corruption operators, however, are applied locally and cannot have been seen.

## 5 Results

### 5.1 Pre-registered verdict: the strong prediction is refuted

Table 1 reports severity-5 detection for all canonical runs. Criterion C2 — entropy near chance — fails on *every* corruption (minimum severity-5 AUROC( $\mathcal{H}$ ) = 0.462, maximum 0.888). The conjunction  $C1 \wedge C2 \wedge C3$  is therefore refuted in all configurations, as recorded in each run’s verdict block (`prediction_confirmed: false`). Realistic corruptions do not translate the energy landscape uniformly: they also reshape relative distances, which entropy sees.

### 5.2 Free energy is a signed detector

The refutation has a clean geometry. On structural corruptions the corrupted embeddings move *away* from all text anchors: mean  $F$  rises monotonically with severity (BreastMNIST speckle:  $1.288 \rightarrow 1.351$ ; pixelate:  $1.288 \rightarrow 1.479$ ) and  $F$  dominates  $\mathcal{H}$  significantly (Table 1, upper block; all DeLong  $p < 0.05$ , CIs excluding zero). On photometric corruptions the embeddings move *closer* to the anchors — mean  $F$  *decreases* (contrast\_down:  $1.288 \rightarrow 1.266$ ; PneumoniaMNIST gaussian\_noise:  $1.264 \rightarrow 1.234$ ) — and the one-sided AUROC falls below 0.5, to 0.127 in the extreme case. A plausible reading is that brightness/contrast compression and moderate X-ray noise push images toward the generic, high-density region of CLIP’s image manifold, reducing the image–text modality gap; we do not claim a mechanism beyond the observed sign.

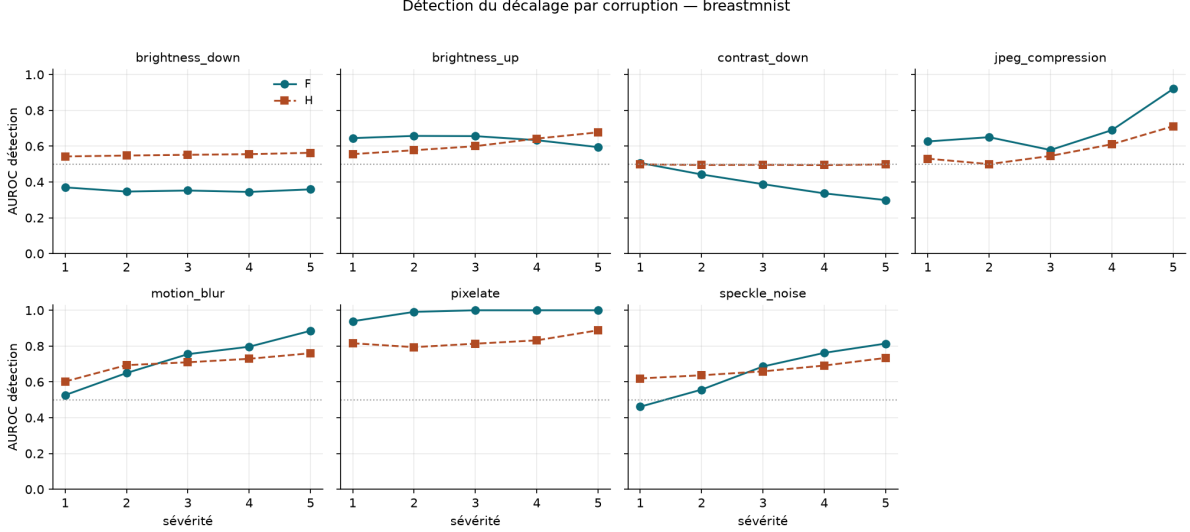


Figure 1: Severity-resolved detection AUROC of  $F$  (solid) and  $\mathcal{H}$  (dashed) for the seven official BreastMNIST-C corruptions. Dotted line: chance. Note the below-chance trajectories of  $F$  on `brightness_down` and `contrast_down`.

Two practical consequences follow. First, using raw  $F$  (or its negation, as in standard energy-score OOD pipelines) as a shift alarm silently fails on exactly the photometric drifts most typical of scanner and probe settings. Second, the failure is directional, hence recoverable.

### 5.3 A source-calibrated two-sided score rescues the inversions

The statistic  $|F - \text{med}_{\text{src}}(F)|$  uses only source images. At severity 5 it restores above-chance detection on five of the six sign-inverted cases — PneumoniaMNIST `gaussian_noise`  $0.127 \rightarrow 0.803$ , BreastMNIST `contrast_down`  $0.299 \rightarrow 0.630$ , `brightness_down`  $0.359 \rightarrow 0.606$ , PneumoniaMNIST `brightness_down`  $0.200 \rightarrow 0.598$  and `contrast_down`  $0.209 \rightarrow 0.589$  — while leaving structural cases essentially unchanged (`pixelate` 1.000, `jpeg` 0.899, `motion_blur` 0.845, `gaussian_blur` 0.962). The recovery is strong where the downward drift is large (X-ray Gaussian noise) and modest where the drift is shallow relative to source dispersion (X-ray photometric shifts,  $\approx 0.59$ , where  $\mathcal{H}$  itself reaches 0.64–0.70). The residual failure, `brightness_up` (0.447), is a genuinely hard case: its  $F$  trajectory is non-monotonic (up at low severity, down at high), so no function of  $F$  alone can order it; its entropy signal (0.677) and the X-ray photometric cases together suggest a bivariate  $(F, \mathcal{H})$  score as the natural next step.

### 5.4 Prompt sensitivity: $F$ is more stable than $\mathcal{H}$

The anchors are literally defined by prompt wording, so prompt sensitivity is the substrate-specific confound. Re-running the full protocol with each single template (BreastMNIST, severity 5):

| Corruption                   | AUROC( $F$ ) |       |       |       | AUROC( $\mathcal{H}$ ) |       |       |       |
|------------------------------|--------------|-------|-------|-------|------------------------|-------|-------|-------|
|                              | t0           | t1    | t2    | ens.  | t0                     | t1    | t2    | ens.  |
| <code>speckle_noise</code>   | 0.778        | 0.801 | 0.854 | 0.814 | 0.777                  | 0.641 | 0.597 | 0.734 |
| <code>brightness_down</code> | 0.351        | 0.419 | 0.307 | 0.359 | 0.580                  | 0.464 | 0.609 | 0.563 |

The dispersion of  $F$  across templates (s.d. 0.032 on `speckle`, 0.046 on `brightness_down`) is roughly half that of  $\mathcal{H}$  (0.076 and 0.064): the partition function aggregates the anchor geometry

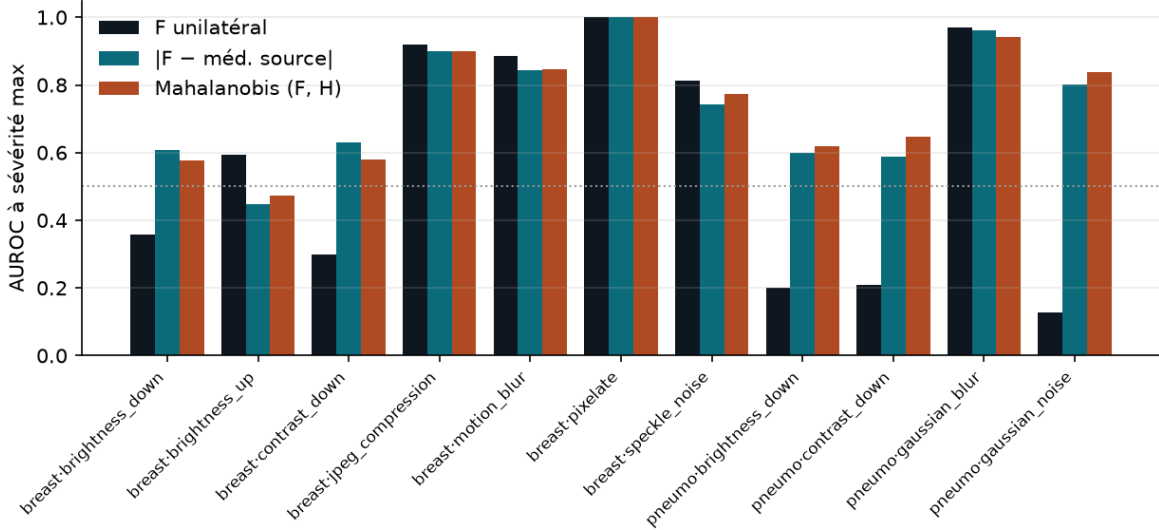


Figure 2: One-sided  $F$  vs. two-sided  $|F - \text{med}_{\text{src}}(F)|$  detection AUROC at maximum severity, all canonical runs.

and is less exposed to a particular wording than the posterior shape. The sign inversion on brightness\_down persists under *every* template — it is not a prompt artifact.

### 5.5 Robustness to corruption sampling

Corruption operators are stochastic. Across three seeds (BreastMNIST, speckle\_noise, severity 5):  $\text{AUROC}(F) = 0.814/0.819/0.827$ ,  $\text{AUROC}(\mathcal{H}) = 0.734/0.749/0.738$ ; the paired severity-5  $\Delta\text{AUROC}$  ranges  $+0.069$  to  $+0.088$  with overlapping CIs. Sampling noise is an order of magnitude smaller than the effects reported above.

### 5.6 Second modality

The chest X-ray cohort replicates the dichotomy on an independent modality, anatomy and corruption implementation, with  $4\times$  larger test sets ( $n = 624$ ). Gaussian blur (structural) behaves like the ultrasound structural block:  $F$  rises monotonically ( $1.264 \rightarrow 1.320$ ) and dominates  $\mathcal{H}$  ( $0.970$  vs.  $0.798$ ,  $\Delta = +0.172$ ,  $p = 3.9 \times 10^{-41}$ ). All three photometric corruptions invert the sign of  $F$  — Gaussian noise, brightness and contrast compression each move X-ray embeddings *closer* to the text anchors ( $F$  AUROC  $0.127$ – $0.209$ ), with effect sizes far beyond sampling noise ( $|\Delta| \geq 0.335$ ,  $p \leq 10^{-86}$ ). On this modality the inversions are deeper than in ultrasound, and  $\mathcal{H}$  is the better detector for the two photometric-compression cases ( $0.644$  and  $0.698$ ): neither observable subsumes the other, which is precisely the argument for reporting both and for the bivariate score sketched above.

## 6 Discussion and limitations

**What the refutation teaches.** The pre-registered conjecture assumed acquisition shift acts as a quasi-uniform translation of the energy landscape. The data say otherwise: realistic corruptions move embeddings along the modality-gap axis in *either* direction and simultaneously reshape relative distances. The corrected picture —  $F$  as a signed coordinate of drift along the anchor-distance axis,  $\mathcal{H}$  as a shape statistic — is more useful than the original conjecture, and directly actionable via the two-sided score.

**Implications for test-time adaptation.** Entropy-minimizing TTA (Wang et al., 2021; Shu et al., 2022) adapts wherever entropy is high. Our results exhibit shifts where entropy barely moves while  $F$  moves strongly (JPEG:  $\mathcal{H}$  AUROC 0.711 vs.  $F$  0.920) and shifts where entropy rises while  $F$  falls. A gating rule based on  $|F - \text{med}_{\text{src}}|$  is the zero-cost candidate this study puts forward for the adaptation-triggering and early-stopping stages of our broader research program on frozen vision–language encoders in ultrasound.

**Limitations.** (i) Test cohorts are small ( $n = 156$  ultrasound images); CIs are reported accordingly. (ii) Single backbone (CLIP ViT-B/16, OpenAI weights); a biomedical encoder (Zhang et al., 2023) may shift the geometry and is the designated robustness check. (iii) MedMNIST-C corruptions are synthetic surrogates for acquisition shift; the multi-center ultrasound transfer (BUSI→UDIAT→BUS-UCLM) is the pre-registered next cohort. (iv) The two-sided median is computed on the full source cohort (in-sample); with  $n = 156$  and one degree of freedom the optimism is negligible, but a held-out calibration split is the clean design. (v) Images are  $224 \times 224$  upsamplings of MedMNIST sources. (vi) CLIP pre-training contamination cannot be audited (Section 4).

## 7 Conclusion

Under a zero-training, leakage-controlled, pre-registered protocol on official medical corruption benchmarks, the Helmholtz free energy of a frozen CLIP posterior is a strong but *signed* detector of acquisition shift: significantly better than entropy on structural corruptions, below chance on photometric ones, and repaired in five of six failure cases by a two-sided source-calibrated variant that requires nothing but source images. The strong form of our pre-registered prediction — entropy at chance — is refuted and reported as such. Every number in this paper is traceable to a versioned run; code, metrics and raw scores are released with the paper.

**Reproducibility.** All experiments: single CPU, no training;  $\sim 3$  minutes per corruption sweep (156 images  $\times$  6 severities). Code, per-run `metrics.json`, raw scores and analysis scripts are available in the project repository.

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